الأولمبياد الوطني للإبداع العلمي
National Olympiad for Scientific Creativity

Developing an Extended Version of the Current National Pre-Marital Screening Programme to Cover Genetic Disorders Associated with Cardiovascular Diseases among Saudis

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Abstract

Problem statement

In March 2022, The National Health Centre, Saudi Health Council, announced that cardiovascular diseases (CVD) are a public health priority. The CVD

Project purpose

This project aims to develop an extended version of the current National Premarital Screening Programme (NPSP), which includes an extra laboratory test designed to detect genetic disorders associated with increased risk of common CVD among Saudis.

Procedure

First, Literature was reviewed to identify common CVD among Saudis. Secondly, information about genetic disorders associated with risks for CVD identified at first was gathered and analysed. Finally, genetic disorders identified in the second step were minimised to an optimum number that allow them to fit the screening programme appropriately.

Results

The extended version of NPSP includes 8888 8888 8888 and 8888, as additional tests that detect three CVDs identified as common CVDs or associated health conditions with increased high risk for developing CVDs.

Project implications

ملخص المشروع

سؤال البحث وأهميته

الغرض من المشروع

الفرضيات التي تم اختبارها

الإجراء المتبع وتحليل البيانات

الاستنتاج

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1. Introduction:

In 2004, Saudi Arabia initiated the National Premarital Screening Programme (NPSP) titled Healthy Marriage, aiming to screen the marriage partners prior to marriage for common genetic disorders (thalassemia and sickle cell anaemia) and infectious diseases (hepatitis B, hepatitis C, and HIV/AIDS). In addition, another similar screening programme was established in 2005, this programme targeting screening neonatal (National New-born screening programme - NSNP) for genetical disorders such as metabolism and endocrine disorders, in 2016, the programme was extended to include screening for hearing loss and critical congenital heart defects. These screening programmes The NPSP and NSNP aid to achieve goals of the Saudi Vision 2030 – (strategic plan; 2.1.3), which aims to promoting prevention health risks, by early detect of individuals who are affected or carries genetic disorders that may lead to hereditary diseases among future generations.

Early identifying of population individuals affected with genetic disorders or carrying these types of disorders and may transfer it to their children resulting in making new affected individuals, by applying screening programme on targeted population, help in reduce the number of new affected individuals, and consequently improves public health. According to World Health Organisation (WHO) Non-communicable disease (NCD) causes count for 71% of premature death cases, of these 30% cases were died due to cardiovascular diseases (CVD), and 13% of cases were died due to cancer, In Saudi Arabia, NCD causes death in 73% of all cases, CVD account for 42% and cancer 7%. Occurring of NCD in high incidence rate among a population affect both public health and healthcare system by; i) affecting individuals' productivity and increasing absenteeism and/or presentism of individuals due to either the disease itself and/or consequences resulted from the disease such as emotional wellbeing, or even responsibility of an individual to take care of an affected related-person, ii) Studies indicate that more than one thirds of healthcare systems resources were spent on patients with genetic disorders, chromosomal disorders, or congenital disorders. The consequences from genetic disorders and its resulted disease can be minimized by early detection of the disease occurring among neoborn or defining individuals carrying these genetic disorders and can transfer it to new individuals of the population. In Saudi Arabia, current NPSP found to be associated with markedly reducing prevalence of hemoglobinopathies, in addition, studies indicate that applying NPSP aided in reducing individuals with risky marriage by 17% from proceeding with marriage in period between 2005 and 2009. The preventive role of the NPSP in reducing risk for existence of individual with genetic disorders, credited to promoting individuals and population awareness of consequences of genetic disorders and hereditary disease and what would be resulted from marrying of couples known to carry or affected by hereditary disease. Searching literature showed limited number of studies that evaluate the advantages of applying screening programmes thus further effort must be focusing on this area of research.

Owing to the high risk of hereditary disease among Saudis, which believed to be a result of practicing consanguineous marriage, NPSP gained a high weight in defining individuals intended to marry but having risks to have in their offspring an affected or carrying genetic disorders person. Consanguinity is a common practice among Arab and Muslims country, in Saudi Arabia, prevalence of consanguinity ranged between 40% and 60% among different cities, whilst 1st cousin marriage may reach 70% of consanguineous marriage in some cities, offspring from this type of marriage known to be related to incidence of many hereditary disease's sech as CVD, cancer, congenital heart disease, thalassaemia, cystic fibrosis, Down's syndrome, infantile cerebral palsy, and hearing and visual disabilities.

Several aspects could be considered to maximize benefits from applying NPSP and minimize incidence of diseases may result from relation such as consanguineous marriage or individuals with known risk to transfer diseases to their children; these aspects including, i) promoting awareness of the population about the importance of NPSP and why this tool is important in reducing risks for developing genetic disorders, ii) oblige/encouraging individuals tend to marry but having high risks to have children affected or carrying serious genetic disorder not to proceed with marriage. iii) updated current NPSP to involve additional panel of test that detect most common disease that affecting the public health of Saudis, including serious disease that associated with consanguineous marriage.

Hereditary disease (HD) is defined as the disease that occurs in individuals as a result of inheriting genetic disorders, there are two types of genetic disorders autosomal dominants and autosomal recessives disorders, genetic disorders can be occurred as a result of either chromosomal, multifactorial or monogenic disorders. The genetic disorders can be inherited from parents to their children, and causes disease such as breast, colorectal, prostate, lung cancer and hemoglobinopathies, intellectual disabilities, congenital glaucoma, ciliopathies, inborn errors of metabolism disorders, retinal dystrophies, hearing loss and primary microcephaly. In Saudi Arabia, previous studies showed that common hereditary disease are including hemoglobinopathies, intellectual disabilities, congenital glaucoma, ciliopathies, inborn errors of metabolism disorders, retinal dystrophies, hearing loss and primary microcephaly, these disorders may result from high prevalence of consanguinity among Saudis and/or specific disorders associated with sub-group of the population (trip/region). A cross-sectional study on consanguinity and major genetic disorders in Saudi children indicate that consanguinity markedly associated with increased risk of congenital heart disease (CHD) by 70% among children of consanguineous marriage. The estimated rate for CHD prevalence among Saudis was 14.8 per 1000 births. Studies reported that the common CHD among Saudis were ventricular septal defect (VSD) with prevalence range of 24% to 39.5% followed by atrial septum defect (ASD) which ranged between 8.9% to 11.5%.

Cardiovascular disease is a general term for diseases that affect heart and blood vessels, and usually classified into six categories, namely, coronary heart disease, cerebrovascular disease, peripheral arterial disease, rheumatic heart disease, CHD, deep vein thrombosis and pulmonary embolism. Reviewing literature indicates overall prevalence of CVD in Saudi Arabia is

30% of adult (aged 18 years and above), there is lack for studies that consider prevalence and incidence of different types of CVDs among Saudis. Several genetic disorders were associated with CVD and reported in literature, however, missing information about prevalence of different type of CVD among Saudis make it difficult for researcher to develop a list of genetic disorder targeting our population to imply this list in early detection of individuals with high risks.

Familial hypercholesterolemia is an autosomal dominant hereditary disease that is associated with consanguinity and led if untreated to atherosclerosis and pre-mature CVD, prevalence of this disease seems to be unestablished among Saudis. FH reported to result from a mutation in low-density lipoprotein receptor (LDLR), apolipoprotein B (APOB) or proprotein convertase subtilisin/kexin 9 (PCSK9). A very rare FH cases can be resulted from Autosomal recessive hypercholesterolemia, lipoprotein adaptor protein 1 (LDLAP1) gene. In Arabian gulf countries, including Saudi Arabia, the prevalence of FH is 1:112 which is three times higher than global prevalence.

In summary, studies highlighted the association between genetic disorders and prevalence of CVD, CAD and CHD among Saudi population, however, fewer studies and research effort were considered the association between genetic disorders and CVD disease phenotyping. Factors associated with increased risks for genetic disorders and hereditary disease among our population are inbreeding (consanguinity) and specific gene disorders identified from sub-groups of the population. Incidence and prevalence of hereditary disease can be reduced by early detection of affected cases as well as identifying individuals carrying autosomal receive gene which may transfer these gene to their children. National screening programme, to identify individuals affected/carrying serious genetic disorders, especially those genetic disorders associated with CVD, must be implemented immediately as a preventive measure, to avoid affecting public health, and to achieve aim of the Saudi Vision 2030, by extending life expectancy of Saudis from 74 to 80.

2. Aim and objectives:

This study aims to develop an extended version of the current national pre-marital screening programme, by introducing a set of laboratory tests that detect genetic disorders associated with the development of common CVD among Saudis, in accordance with public health recommendation (No. 1) included in letter issued by Saudi Health Council (Ref. 32665-1, 15-09-2022), that urge all interested, expertise, specialist and healthcare providers to consider the CVD as a national health priority.

Objectives:

a) .

3. Materials And Methods:

Study design:

This study is designed to review evidence literature for previous findings from literature to select the most appropriate laboratory tests to use in developing an extended version of the current NPSP.

Search strategies:

Scientific database search, namely, PubMed and Google Scholar, was carried to review literature for previous studies that consider CVD among Saudis, using a combination of pre-selected keywords: "cardiovascular ", "Saudi", "coronary heart disease", "Familial hypercholesterolemia", " Mendelian disorder", "Arab", "Gulf countries", "genetic disorders", "autosomal dominants", "autosomal recessive" and "myocardial infraction".

Resources:

All resources required for this study including accessing to scientific database were available online and accessible via authorization given by the King Abdulaziz and his Companions Foundation for Giftedness and Creativity "Mawhiba".

Procedures:

To conduct this study, the following steps will be taken:

a) Evaluate the efficiency of current national pre-marital screening programme, by examining evidence reported from previous studies about efficiency of applying the current programme in reducing incidence of hereditary disease.
b) Review literature to define the list of common hereditary disease among Saudis that result from genetic disorders.
c) Define possible genetical disorders associated with each of the hereditary diseases selected at (b)
d) Defining possible laboratory tests that can detect genetical disorders selected at (c) and evaluate most cost efficient and accurate laboratory tests that could be introduced to the national pre-marital screening programme.
e) Optimize a new panel of tests to introduce to the current pre-marital screening programme considering the cost-efficiency, accuracy, reliability, and applicability of the panel.

Ethics considerations:

This research does not need ethical approval as it is not included any personal information or experiments that involves any human or animal.

4. Result

In accordance with the study purposes, Findings from this study organised in three sections as follow:

Prevalence of cardiovascular disease among Saudis

Reviewing previous studies that designed to estimate the prevalence of common CVD among Saudis, indicates that 8888 8888 88888 8888 8888 were most prevalent diseases of CVD among our population. **Table 1**, summarises the findings from previous studies that reported prevalence of the CVD among Saudis.

Table 1. Reported prevalence of the CVD among Saudis.

No	Disorders	Prevalence	References	Year
1				
2				
3				
4				
5				
6				
7				
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9				
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11				
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13				
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15				
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Genetic disorders associated with common cardiovascular disease among Saudis

Table 2. cardiovascular diseases and associated genetic disorders.

No	Disease	Associated genetic disorders

Developed version of the current National Premarital Screening Programme (NPSP)

Based on evidence highlighted earlier there are an urgent need to include test for 888 8 8 8 8 8

Within the NPSP, thus the final version of the updated national premarital screening programme will be including the following tests.

a) Viral Infection: Hepatitis B, Hepatitis C and Human Immune Deficiency (HIV)/AID.
b) Blood Disorders: Thalassemia and Sickle cell anaemia.
c) Cardiovascular diseases: 88888 88 8 8 888 888 8 8 8 8 8 8

Cardiovascular diseases and associated risks factors can be screened using genetic disorders summarised in **Table 3**.

Table 3. Summary of cardiovascular diseases and associated genatic disorders.

No	Disease	Associated genetic disorders
1		
2		
3	Familial hypercholesterolemia (FH)	Low-density lipoprotein receptor (LDLR) Apolipoprotein B (APOB), Proprotein convertase subtilisin/kexin 9 (PCSK9), Lipoprotein adaptor protein 1 (LDLAP1)
4		
5		

5. Conclusion:

Cardiovascular diseases have been recently announced as a national health priority, due to serious consequences of these type of diseases that affect public health. Existence of high prevalence of consanguinity among our population emphasis the needs to investigate role of inherent genetic disorders in risk for developing CVD. Furthermore, in absence of precise estimate for prevalence of CVD and its associated risks factors among Saudis, the extended version of the NPSP developed in this study, will provide a reliable screening tool, identify current prevalence of CVD diseases resulted from genetic disorders among our population, in addition to identify risk for inherent of autosomal recessive disorders related to CVD by marriage to future generations.

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6. References: